

Exercise 7-2: Prepare the Cars data

Did you use a language model?

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In [ ]: ### A : No, I did not find any use for a language model for this assignment exce
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Read the data

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In [ ]: # Import the pandas library  
import pandas as pd
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In [ ]: # Load the dataset from a CSV file named 'cars.csv' into a DataFrame named 'cars'  
cars = pd.read_csv('cars.csv')
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In [ ]: # Display the first 5 rows of the 'cars' DataFrame to provide an overview of the  
cars.head()
```

Add and drop columns

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In [ ]: # Retrieve and display the unique values in the 'CarName' column, which may incl  
cars.CarName.unique()
```

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In [ ]: # Create a new column 'brand' by applying a lambda function to each row. This fu  
cars['brand'] = cars.apply(lambda x: x.CarName.split(' ')[0], axis=1)
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In [ ]: # Display the unique values in the 'brand' column to verify the extraction of ca  
cars.brand
```

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In [ ]: # Define a function 'get_model' to extract the model name from the 'CarName' col  
def get_model(row):  
    names = row.CarName.split(' ') # split into a list of names  
    name = ' '.join(names[1:]) # join all names except the first, which is  
    return name  
    # Apply the 'get_model' function to each row to create a new 'model' column  
cars['model'] = cars.apply(get_model, axis=1)  
  
# Display the unique values in the 'model' column to verify the extraction of ca  
cars.model
```

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In [ ]: # Remove the columns 'car_ID' and 'CarName' from the DataFrame, as they're no Lo  
cars.drop(columns=['car_ID', 'CarName'], inplace=True)
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In [ ]: # Display the first 5 rows of the modified 'cars' DataFrame to see the results o  
cars.head()
```

Set an index and unstack the data

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In [ ]: # Create a new DataFrame 'cars_filtered' by filtering 'cars' for rows where the
cars_filtered = cars.query('brand == "volkswagen"')[['model', 'horsepower', 'carbo

# Display the first 5 rows of 'cars_filtered' to inspect the filtered and select
cars_filtered.head()
```

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In [ ]: # Set a MultiIndex for 'cars_filtered' using the 'model', 'carbody', and 'doornu
cars_indexed = cars_filtered.set_index(['model', 'carbody', 'doornumber'])

# Display the 'cars_indexed' DataFrame to see the structure with the newly set M
cars_indexed
```

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In [ ]: # Unstack the 'cars_indexed' DataFrame by the 'carbody' level of the index, rear
cars_indexed.unstack('carbody')
```